POWER PULSE

Transmission Line Performance Important Formulae

Notes by Balaji Sir

1 ABCD constants

1. $$V_s = AV_r + BI_r, \quad I_s = CV_r + DI_r, \quad AD - BC = 1$$

for a passive reciprocal network

2. $$\text{Short: } A = D = 1,\ B = Z,\ C = 0$$

3. $$\text{Nominal-}\pi\text{: } A = D = 1 + \frac{ZY}{2},\ B = Z,\ C = Y\left(1 + \frac{ZY}{4}\right)$$

4. $$\text{Nominal-T: } A = D = 1 + \frac{ZY}{2},\ B = Z\left(1 + \frac{ZY}{4}\right),\ C = Y$$

5. $$\text{Long: } A = D = \cosh \gamma l,\ B = Z_c \sinh \gamma l,\ C = \frac{\sinh \gamma l}{Z_c}$$

y = √(zy), Z_c = √(z/y)

6. $$\text{Lossless: } A = D = \cos \beta l,\ B = jZ_c \sin \beta l,\ C = \frac{j \sin \beta l}{Z_c}$$

β = ω√(LC)

2 Regulation, efficiency, Ferranti

7. $$\%\,\text{Reg} = \frac{|V_{r,NL}| - |V_{r,FL}|}{|V_{r,FL}|} \times 100, \quad |V_{r,NL}| = \frac{|V_s|}{|A|}$$

8. $$\Delta V \approx I\,(R\cos\phi \pm X \sin\phi)$$

+ lagging pf, − leading pf

9. $$\eta = \frac{P_r}{P_s} \times 100$$

10. $$V_r = \frac{V_s}{\cos \beta l} \qquad \text{(open-ended lossless line)}$$

Ferranti effect ∝ l^2

3 Combinations

11. $$\text{Cascade: } \begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} A_1 & B_1 \\ C_1 & D_1 \end{bmatrix} \begin{bmatrix} A_2 & B_2 \\ C_2 & D_2 \end{bmatrix}$$

12. $$\text{Parallel: } A = \frac{A_1B_2 + A_2B_1}{B_1 + B_2},\ B = \frac{B_1B_2}{B_1 + B_2}$$

13. $$\lambda = \frac{2\pi}{\beta}, \qquad v = \frac{1}{\sqrt{LC}}$$

wavelength and velocity